

# Track 2H-L Latent-State Hysteresis Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

## Overview

This report closes the approved `Track 2H-L` causal latent-state / hysteresis-aware campaign; all six planned runs completed, and the program-level scalar winner did not change.

## Campaign Artifacts

| Artifact         | Path                                                                                                                                        |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Campaign output  | <code>output/training_campaigns/2026-06-16-18-06-11_track2h_latent_state_hysteresis_campaign_2026_06_16</code>                              |
| Leaderboard      | <code>output/training_campaigns/2026-06-16-18-06-11_track2h_latent_state_hysteresis_campaign_2026_06_16/campaign_leaderboard.yaml</code>    |
| Best-run pointer | <code>output/training_campaigns/2026-06-16-18-06-11_track2h_latent_state_hysteresis_campaign_2026_06_16/campaign_best_run.yaml</code>       |
| Execution report | <code>output/training_campaigns/2026-06-16-18-06-11_track2h_latent_state_hysteresis_campaign_2026_06_16/campaign_execution_report.md</code> |
| Planning report  | <code>doc/reports/campaign_plans/track2/2026-06-16-16-00-57_track2h_latent_state_hysteresis_campaign_plan_report.md</code>                  |
| Campaign package | <code>config/training/track2h_latent_state_hysteresis/campaigns/2026-06-16_track2h_latent_state_hysteresis_campaign</code>                  |

## Execution Summary

| Field                               | Value                                                            |
|-------------------------------------|------------------------------------------------------------------|
| Campaign name                       | <code>track2h_latent_state_hysteresis_campaign_2026_06_16</code> |
| Completed runs                      | 6                                                                |
| Failed runs                         | 0                                                                |
| Model type                          | <code>latent_state_hysteresis_probe</code>                       |
| Profiles                            | <code>gru</code> , <code>tcn</code>                              |
| Tested surfaces                     | <code>global</code> , <code>Fw</code> , <code>Bw</code>          |
| Runner-level scalar first entry     | <code>te_track2h_l_causal_tcn_offset_residual_global</code>      |
| Program-level scalar winner changed | no                                                               |

## Directional Branch Results

| Surface | Candidate  | Profile | Encoder | Test MAE | Test RMSE | Val MAE  |
|---------|------------|---------|---------|----------|-----------|----------|
| global  | tcn_global | tcn     | C-TCN   | 0.003368 | 0.003860  | 0.003543 |
| Fw      | tcn_fw     | tcn     | C-TCN   | 0.003470 | 0.004068  | 0.003565 |
| Bw      | gru_bw     | gru     | GRU     | 0.003545 | 0.004175  | 0.003837 |

These rows are the best scalar result per branch. The global causal-TCN entry is the campaign scalar leader, but it does not replace the forward or backward branch decisions.

## Latent-State Leaderboard

| Rank | Surface | Candidate  | Profile | Encoder | Test MAE | Test RMSE | Val MAE  |
|------|---------|------------|---------|---------|----------|-----------|----------|
| 1    | global  | tcn_global | tcn     | C-TCN   | 0.003368 | 0.003860  | 0.003543 |
| 2    | Fw      | tcn_fw     | tcn     | C-TCN   | 0.003470 | 0.004068  | 0.003565 |
| 3    | Fw      | gru_fw     | gru     | GRU     | 0.003537 | 0.004110  | 0.003468 |
| 4    | Bw      | gru_bw     | gru     | GRU     | 0.003545 | 0.004175  | 0.003837 |
| 5    | global  | gru_global | gru     | GRU     | 0.003590 | 0.004074  | 0.003717 |
| 6    | Bw      | tcn_bw     | tcn     | C-TCN   | 0.003630 | 0.004312  | 0.003840 |

## Encoder Split

| Surface | TCN Candidate | TCN MAE  | GRU Candidate | GRU MAE  | Better |
|---------|---------------|----------|---------------|----------|--------|
| global  | tcn_global    | 0.003368 | gru_global    | 0.003590 | C-TCN  |
| Fw      | tcn_fw        | 0.003470 | gru_fw        | 0.003537 | C-TCN  |
| Bw      | tcn_bw        | 0.003630 | gru_bw        | 0.003545 | GRU    |

The causal-TCN profile is stronger on `global` and `Fw`; the recurrent GRU profile is stronger only on `Bw`. This suggests that short causal-history features help, but a generic hidden-state encoder is not enough by itself to solve the dispersed-offset problem.

## Track 2H Dispersion-Aware Comparison

| Surface | 2H-L   | 2H-L MAE | MDN     | MDN MAE  | Prob.  | Prob. MAE Robust | Robust MAE |          |
|---------|--------|----------|---------|----------|--------|------------------|------------|----------|
| global  | tcn_g  | 0.003368 | mdn2_g  | 0.003503 | nll_g  | 0.003013         | mae_g      | 0.003406 |
| Fw      | tcn_fw | 0.003470 | mdn3_fw | 0.003235 | nll_fw | 0.003165         | mae_fw     | 0.003146 |
| Bw      | gru_bw | 0.003545 | mdn2_bw | 0.002658 | q_bw   | 0.002927         | sll_bw     | 0.003074 |

The latent-state campaign improves the scalar `global` surface versus the MDN and robust-loss `global` baselines, but it remains behind the Gaussian-NLL probabilistic `global` candidate. It is weaker than the existing `Fw` and `Bw` dispersion-aware leaders.

## Registry Effects

The campaign runner refreshed the six family registries and updated the program registry. The program-level scalar best did not move: `te_periodic_gru_sequence_remote_Bw` remains the current scalar program winner with test MAE = 0.002344 .

| Registry Scope                | BestRun                                                     | Test MAE |
|-------------------------------|-------------------------------------------------------------|----------|
| Track 2H-L global TCN         | <code>te_track2h_l_causal_tcn_offset_residual_global</code> | 0.003368 |
| Track 2H-L Fw TCN             | <code>te_track2h_l_causal_tcn_offset_residual_fw</code>     | 0.003470 |
| Track 2H-L Bw GRU             | <code>te_track2h_l_gru_offset_residual_bw</code>            | 0.003545 |
| Current program scalar winner | <code>te_periodic_gru_sequence_remote_Bw</code>             | 0.002344 |

The best `Track 2H-L` branch is about 43.7% worse than the current program scalar winner, so it is not a direct program promotion.

## Track 2 Boundary

Official `Track 2` curve-first verification was not run as part of this normal campaign closeout. Under campaign governance, that remains a separate operator-approved workflow after this campaign-results report and PDF are complete.

If launched, the next `Track 2` refresh should add all six `Track 2H-L` candidates and report `global` , `Fw` , and `Bw` decisions separately, comparing raw error, centered-shape error, offset behavior, amplitude, harmonic behavior, collage plots, and overlays against the completed `Track 2H` , `Wave 3` , `Track 2G` , and accepted direction-parallel baselines.

## Closeout Decision

`Track 2H-L` execution is complete: all planned candidates produced successful training artifacts, no failed run remains, registries were refreshed by the runner, and the active campaign state can be cleared.

From a modeling standpoint

- carry the causal-TCN `global` result forward as useful evidence that causal history helps the global scalar surface;
- do not promote `Track 2H-L` over the current dispersion-aware `Fw` or `Bw` leaders;
- treat GRU hidden state as branch-specific evidence, not as a generally superior hysteresis solution;
- keep latent-state / hysteresis-aware modeling available for later multi-head integration only if curve-first verification shows offset or continuity advantages that scalar MAE does not expose.

## Recommended Follow-Up

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1. Accept this closeout and clear the active campaign state.
2. Optionally prepare a separate operator-launched Track 2 verification refresh for all six Track 2H-L candidates.
3. If the official Track 2 refresh does not show a curve-quality gain, move the roadmap toward Wave 4 / integrated multi-head design using 2H-L only as diagnostic evidence.